Supply Chain Optimization

Multi-item control

Service level differentiation

Overview

- ABC classification and service levels
- Service level definition
- SKU service levels

APICS

- ABC classifies of a group of SKUs in decreasing order of annual dollar volume (price multiplied by projected volume) or other criteria. This array is then split into three classes, called A, B and C.

- Typically, A contains 20% of all SKUs, B 30%, and C 50%.

Example

SKU	Weekly Demand	Cost Price
1	100	10
2	100	1
3	20	100
4	20	10
5	20	1
6	5	100
7	5	50
8	5	20
9	5	10
10	5	1

ABC Demand * Price

SKU	Demand	Price	*Demand * Price*	*Class*
3	20	100	2000	A
1	100	10	1000	A
6	5	100	500	B
7	5	50	250	B
4	20	10	200	B
2	100	1	100	C
8	5	20	100	C
9	5	10	50	C
5	20	1	20	C
10	5	1	5	C

ABC Service levels

- The standard practice is to assign the same service level to each SKU in a particular class.

Xeleos
"A contagious passion for Supply Chain Excellence"

ABC-tool

Hoe uw voorraad onder controle houden

Elk bedrijf wordt de dag van vandaag geconfronteerd met de voorraadproblematiek. Het probleem is tweevoudig. Enerzijds dienen voldoende voorraden als buffer te worden aangelegd om risico en onzekerheid in de vraag en de levertermijn in te dekken. Anderzijds dient de voorraad zo laag mogelijk te worden gehouden omdat dit aanzienlijke kosten en vaak ook organisatorische problemen met zich meebrengt: opslag van goederen vereist ruimte, toezicht, controle, energie, alsook kapitaal. Bovendien kunnen producten in voorraad na enige tijd verouderd, beschadigd, bedorven of simpelweg irrelevant worden.

Ondanks moderne hulpmiddelen zijn in een bedrijf vaak nog onnodig hoge voorraden aanwezig. Deze kunnen te wijten zijn aan een slechte organisatie, een gebrekkige informatie, slechte relaties met de leveranciers, een gebrek aan een efficiënte planning, enz. Een regelmatige grondige analyse van de voorraadsituatie is dus geen overbodige luxe, maar een noodzaak! Een goede voorraad- en bestelpolitiek houdt in dat een optimaal voorraadbeheerssysteem wordt nagestreefd, zodat op elk moment aan de vraag kan worden voldaan, terwijl de totale kosten zo laag mogelijk worden gehouden.

ABC Analyse

Een Pareto analyse – ook ABC analyse genoemd – is een van de meest toegepaste technieken in het voorraadbeheer en heeft als doel inzicht te krijgen in de samenstelling van de aanwezige voorraden. Het deelt de artikelen op volgens verbruikswaarde en verbruiksfrequentie, zodat onmiddellijk duidelijk wordt welke artikelen de grootste bijdrage leveren in de waarde van de voorraad en de omzet. Het resultaat is een onderverdeling van de producten in verschillende klassen. Per klasse worden de artikelen gekenmerkt door specifieke eigenschappen: gemiddelde verbruikswaarde, verbruiksfrequentie, kostprijs, ... Maar is deze analyse diepgaand genoeg?

A 99%
B 92%
C 65%

YOUR COMPANY
CUSTOMER
plan
MAKE
DISTRIBUTE
SUPPLIER
SUPPLY
collaborate
supporting processes

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Class service levels?

- Some authors have argued that A items are the most critical for a firm.
- Others have claimed that dealing with stock-outs is not worth the effort for C items.

Research paper 1

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ABC Classification: Service Levels and Inventory Costs

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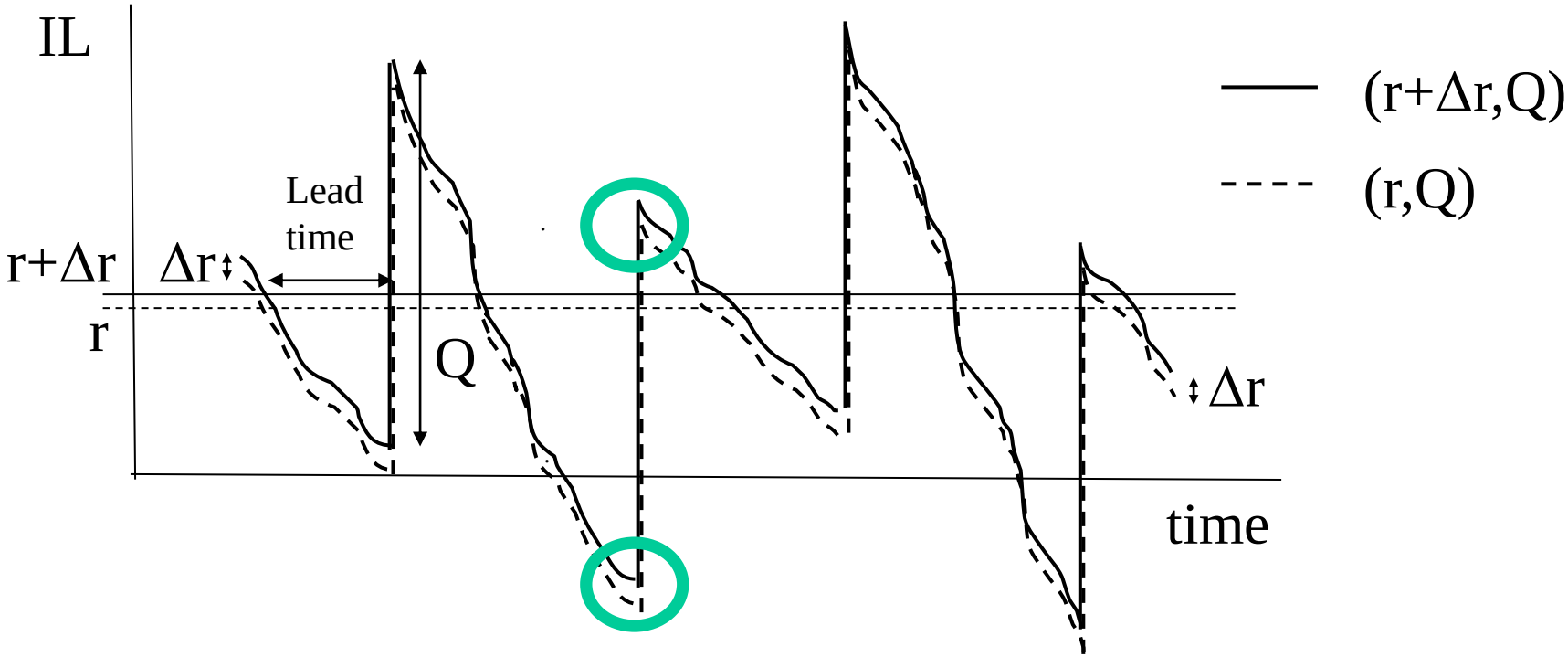
Key observation

- Aiming for the lowest cost under a Fill Rate restriction, implies that
- there is a backorder cost per unit (or lost sale)

- So, referring to the textbook and single item slides, a cost b_2 per unit.

Backorder cost b_2 per unit (General demand)



$$C(r+\Delta r) - C(r)$$
$$= h\Delta r S_2(r) - b_2 \frac{\mu}{Q}\Delta r\left(\left(1 - S_1(r)\right) - \left(1 - S_1(r+Q)\right)\right) + o(\Delta r)$$

In the optimal solution we have approximately (Q large)

$$S_1 \approx 1 - \frac{QhS_2}{b_2\mu} \approx 1 - \frac{Qh}{b_2\mu}$$

NEW ABC classification

- *Rank the SKUs based on the cost criterion* $\frac{Q_i h_i}{b_{2,i}\mu_i}$ *in descending order, make classes, and fix the* S_1 *per class.*

Empirical Investigation

- Dataset 1 (4,799 SKUs) is from a warehouse supplying spare parts globally for the installed base of machines that are used in the textile industry.
- Dataset 2 (39,274 SKUs) is from a retailer that sells bike and car parts and accessories.
- Dataset 3 (9,086 SKUs) is from a retailer that sells do-it-yourself products.

- Three criteria: demand value, demand volume and *cost* (criticality not considered)
- 3 or 6 classes
- 3 classes: A, B and C contains about 20%, 30% and 50% of all SKUs (similar for 6 classes)
- Target overall fill rate s_2 : 95% or 99%
- SAME backorder cost b for all SKUs

- Use Excel to find the s_1 (of at least 50%) per class that minimize the total inventory cost whilst achieving the target overall fill rate.

Dataset 1, Normal demand

		Safety stock cost (% increase over optimal) for ABC based on		
Target Fill rate S_2	No. of classes	Demand * Price	Demand	Cost criterion
95%	3	2041%	1096%	38%
	6	1661%	477%	20%
99%	3	363%	196%	22%
	6	336%	174%	8%

Dataset 2, Normal demand

		Safety stock cost (% increase over optimal)		
Target Fill rate S_2	No. of classes	Demand value	Demand volume	Cost criterion
99%	3	410%	138%	7%
	6	401%	115%	4%

Remark: due to the small lead time of 2 days for all SKUs, a 95% (or in fact a 97%) fill rate is achieved with zero safety stock (i.e. a cycle service level of 50%) for all SKUs.

Dataset 3, Normal demand

		Safety stock cost (% increase over optimal)		
Target Fill rate S_2	No. of classes	Demand value	Demand volume	Cost criterion
95%	3	753%	362%	63%
	6	740%	278%	14%
99%	3	157%	74%	23%
	6	154%	58%	5%

Conclusion

- Standard ABC classification methods, in combination with fixed service levels per class, are far from optimal.
- Using the newly proposed Cost criterion can lead to considerable cost savings.
- Moreover, the new method can take criticality into account.

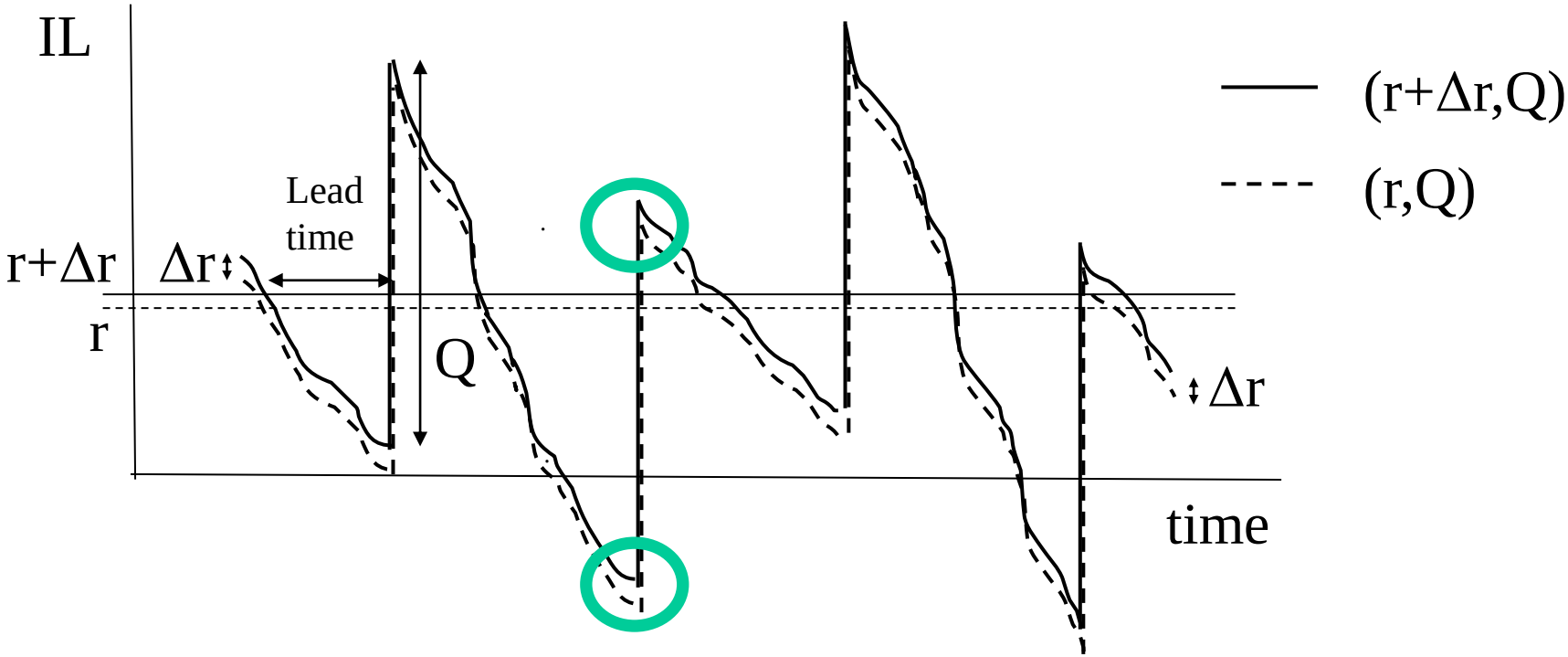
BUT…

- Why classify in the first place?
- The cycle service level S_1 is <u>NOT</u> the service level experienced by the customer!
- The cycle service levels S_1 per class that satisfy a fill rate S_2 constraint can only be determined numerically.

Research paper Unpublished

- R.H. Teunter, A.A. Syntetos, M.Z. Babai, Service Level Differentiation for Assortment Inventory Control, working paper.

Backorder cost b_2 per unit (General demand)



$$C(r+\Delta r) - C(r)$$
$$= h\Delta r S_2(r) - b_2 \frac{\mu}{Q}\Delta r\left(\left(1 - S_1(r)\right) - \left(1 - S_1(r+Q)\right)\right) + o(\Delta r)$$

In the optimal solution we have approximately

$$S_1 \approx 1 - \frac{QhS_2}{b_2\mu} \approx 1 - \frac{QhS_1}{b_2\mu} \quad \text{and so} \quad S_1 \approx 1 - \frac{Qh}{Qh + b_1\mu}$$

- $S_{1,i} \approx 1 - \frac{Q_i h_i}{Q_i h_i + b_1 \mu_i}$ gives $\frac{1 - S_{1.i}}{S_{1,i}} \approx \frac{Q_i h_i}{b_1 \mu_i}$

- Shore (1982) shows that for a standard Normal distribution with loss prob. α, the expected loss is approximately $0.4115 \frac{\alpha}{(1-\alpha)}$.

Shore, H. (1982). Simple Approximations for the Inverse Cumulative Function, the Density Function and the Loss Integral of the Normal Distribution. Appl. Statist, 31 (2), 108-114.

- So the optimal expected loss (backorders) at the end of a cycle is approximately $0.4115\sigma_i' \frac{Q_i h_i}{b_1 \mu_i}$
- To obtain a fill rate of S_2, this also equals $\left(1 - S_{2,i}\right) Q_i$

- Hence, the optimal backorder rate is

$$\left(1 - S_{2,i}\right)\mu_i = \frac{0.4115}{b_1}\sigma_i' h_i$$

Optimal backorder rate division

- So, $\left(1 - S_{2,i}\right)\mu_i = \frac{0.4115}{b_1}\sigma_i' h_i$
- And for all SKUs together, the optimal backorder rate is
- $\sum_i \left(\left(1 - S_{2,i}\right)\mu_i\right) = \frac{0.4115}{b_1}\sum_i(\sigma_i' h_i)$

- So $\frac{\left(1-S_{2,i}\right)\mu_i}{\sum_i\left(\left(1-S_{2,i}\right)\mu_i\right)} = \frac{\sigma_i' h_i}{\sum_i(\sigma_i' h_i)}$

- So $\frac{(1-S_{2,i})\mu_i}{\sum_i\left((1-S_{2,i})\mu_i\right)} = \frac{\sigma_i' h_i}{\sum_i(\sigma_i' h_i)}$

- But the total backorder rate must be $(1 - TFR)\sum_i(\mu_i)$

- So $\left(1 - S_{2,i}\right)\mu_i = \frac{\sigma_i' h_i}{\sum_i(\sigma_i' h_i)}(1 - TFR)\sum_i(\mu_i)$

Minimum fill rate $S_{2,min}$

- However, we must ensure that $S_{2,i} \geq S_{2,min}$
- Or, equivalently, that the backorder rate is at most $\left(1 - S_{2,min}\right)\mu_i$

- This can be done iteratively – see next slide

Iterative procedure

- Order the SKUs from the largest to the smallest value of $\frac{\sigma_i' h_i}{\mu_i}$.
- To initialize, for iteration 1, set the available stock-out rate to $A_1 = (1 - TFR)\sum_i(\mu_i)$.
- Iteration k: Set the fill rate of SKU k to $S_{2,k} = \max\left[1 - \frac{A_k}{\mu_k}\frac{\sigma_k' h_k}{\sum_{j\geq k}\sigma_j' h_j}; S_{2,min}\right]$, set $A_{k+1} = A_k - \mu_k\left(1 - S_{2,k}\right)$, and go to iteration $k + 1$.

Example

Target TFR	95,00%							
FR min	85,00%							
SKU	σ_i'	μ_i	price	holding cost	h_i	$\sigma_i' h_i$	Service Level	Backorders
A	100	150	20	25%	5	500	78,96%	31,56
B	15	560	23	25%	5,75	86,25	99,03%	5,44
C	20	80	35	25%	8,75	175	86,19%	11,05
D	50	250	5	25%	1,25	62,5	98,42%	3,95
sum		1040				823,75		52

Check: TFR=1-52/1040=0,95

SKU A: FR=0,85 gives 0,15*150=22,5 backorders;
52-22,5=29,5 backorders remaining or TFR(BCD)=

Example cont'd

Backorders (CDB)	29,5		TFR(CDB)	96,69%				
FR min	85,00%							
SKU	σ_i'	μ_i	price	holding cost	h_i	$\sigma_i' h_i$	Service Level	Backorders
A	100	150	20	25%	5	500	85,00%	22,5
C	20	80	35	25%	8,75	175	80,07%	15,95
D	50	250	5	25%	1,25	62,5	97,72%	5,69
B	15	560	23	25%	5,75	86,25	98,60%	7,86
sum		890				323,75		52

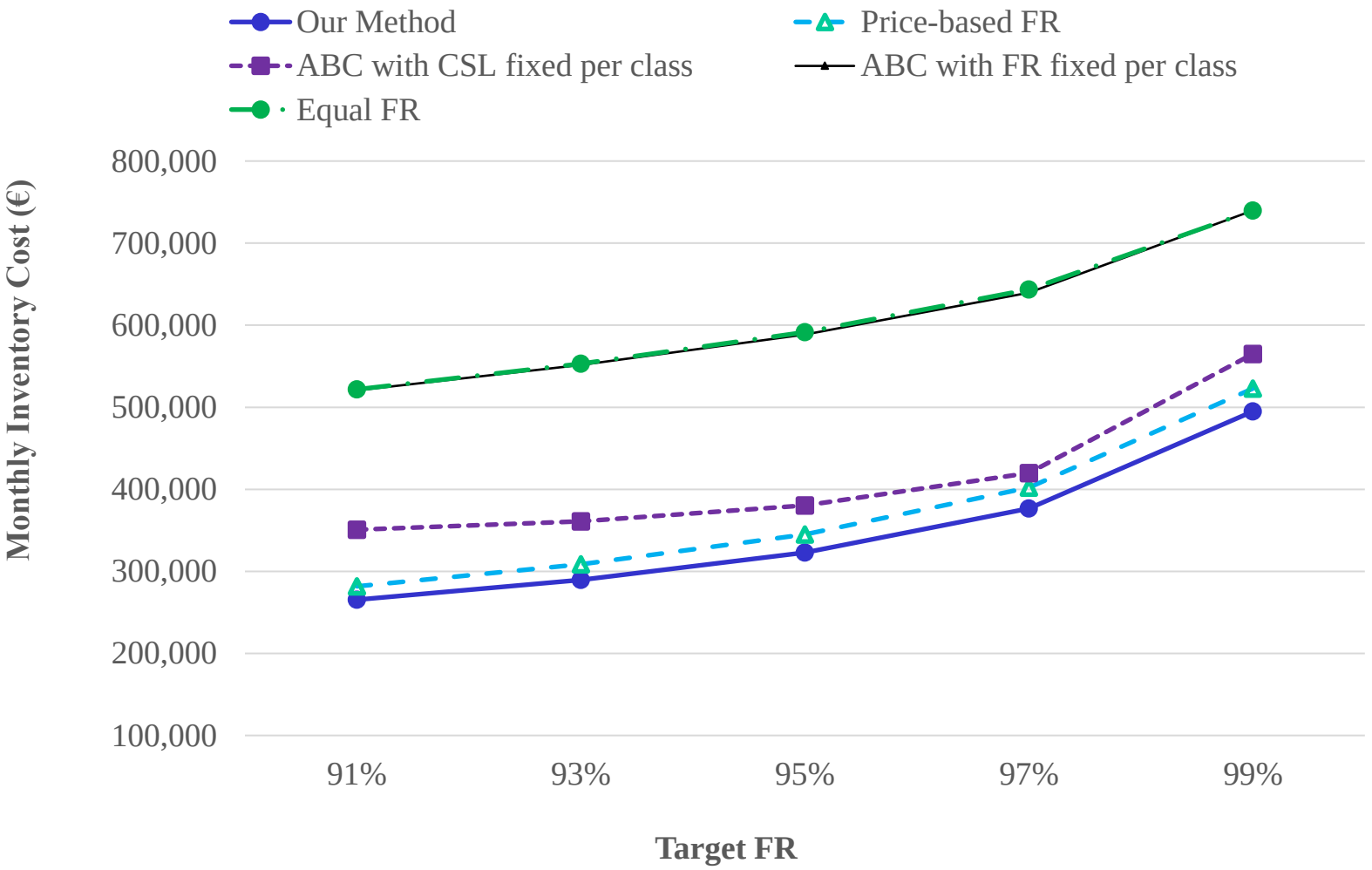
Check: TFR=1-52/1040=0,95

SKU C: FR=0,85 …

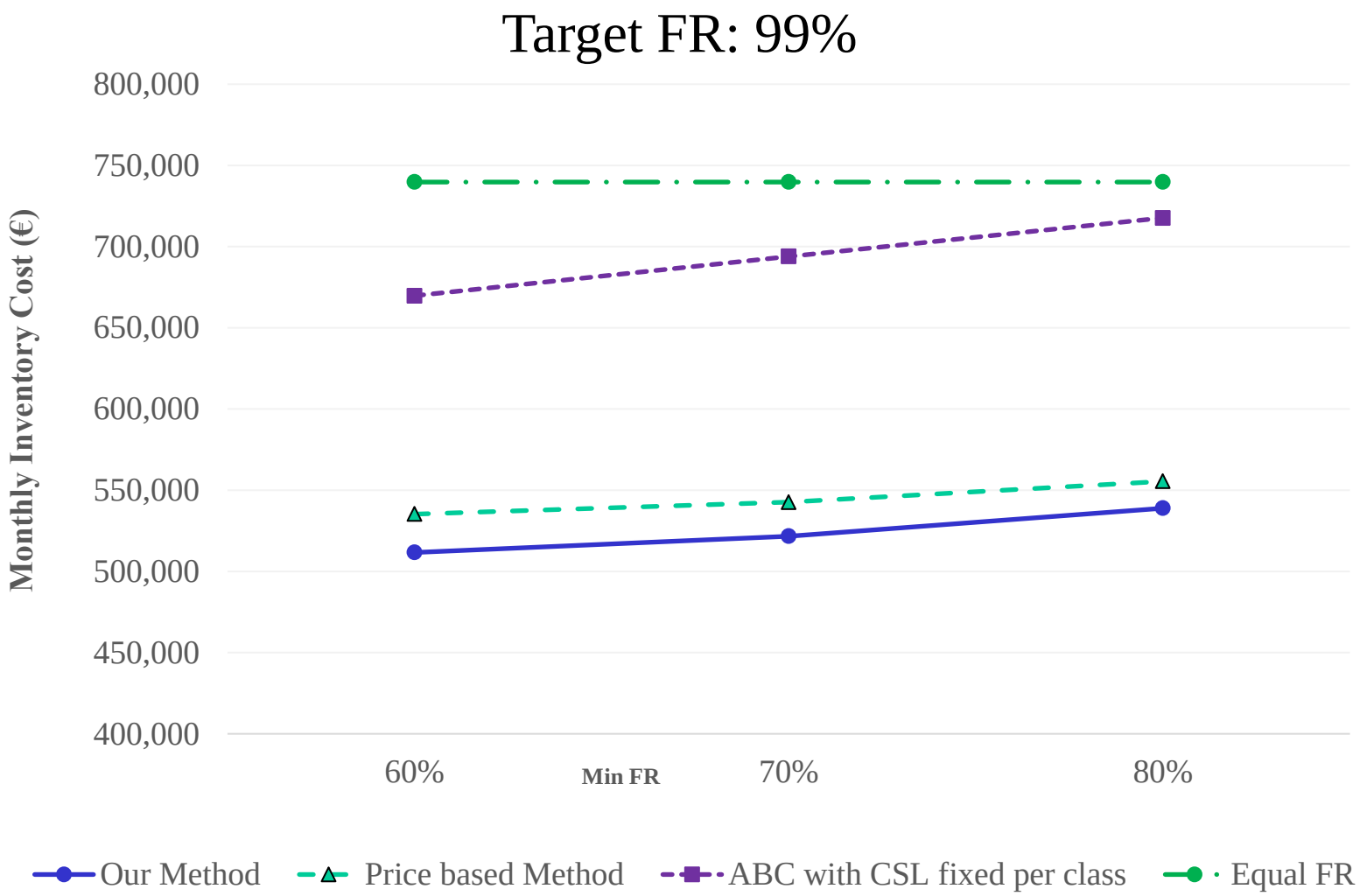
Real life data

- Wholesaler
- Complete: 210k SKUs
- Sample: 5k SKUs

Complete (200k): No minimum FR



Complete(200k)



Sample (5k)
Include Optimal

Backorder penalty: *b*=3

Conclusion

- Any company should implement proper forecasting and SL diffferentiation methods.
- Any inventory course should discuss them!